

## CS 171: Problem Set 1

Due Date: September 11, 2026 at 8.59pm via Gradescope

### 1. Substitute and Shift cipher (10 points)

Consider a modification of the substitution cipher, where instead of applying only the substitution, we first apply a substitution and then apply a shift cipher on the substituted values. Give a formal description of this scheme and show how to break the substitute and shift cipher.

#### Solution

#### Substitution and Shift Cipher.

1. Gen : Choose a random permutation  $f$  of  $\mathbb{Z}_{26}$  and a key  $k$  randomly from  $\mathbb{Z}_{26}$ . The key consists of  $(f, k)$
2. Enc( $(f, k), m$ ) : Output  $c = f(m) + k \pmod{26}$ .
3. Dec( $(f, k), c$ ) : Output  $m = f^{-1}(c - k \pmod{26})$ .

**Attack.** We will show that this scheme is exactly the same as a substitution cipher, but with a different way of sampling the substitution key.

For each key pair  $(f, k)$ , let  $g_{f,k}$  be the permutation applied by the combined cipher:

$$g_{f,k}(x) = f(x) + k \pmod{26}.$$

Since adding  $k$  is a permutation of  $\mathbb{Z}_{26}$ ,  $g_{f,k}$  is itself a permutation. Thus, encrypting with the pair  $(f, k)$  is the same as using  $g_{f,k}$  as the key of an ordinary substitution cipher.

Now fix a substitution key  $g$ . How many pairs  $(f, k)$  produce this same substitution? For each of the 26 choices of  $k$ , there is exactly one possible choice of  $f$ , namely

$$f(x) = g(x) - k \pmod{26}.$$

This  $f$  is a permutation, and it satisfies  $f(x) + k = g(x) \pmod{26}$ . Therefore, exactly 26 key pairs produce each substitution key  $g$ : one pair for each choice of  $k$ .

The scheme chooses uniformly from  $26! \cdot 26$  key pairs. Since exactly 26 of those pairs produce any particular  $g$ ,

$$\Pr[g_{f,k} = g] = \frac{26}{26! \cdot 26} = \frac{1}{26!}.$$

Thus  $g_{f,k}$  is uniform over all  $26!$  permutations. The substitute-and-shift cipher is therefore distributed exactly like the ordinary substitution cipher; it merely samples the substitution permutation in a different way. The same frequency-analysis attack consequently breaks it.

## 2. Double Vigenère cipher (10 points)

Consider the double Vigenère cipher, where we choose two keys  $k_1, k_2$  with different periods  $t_1, t_2 \in [t_{max}]$ , and then encrypt a message in the following way: first encrypt the message using the Vigenère cipher with  $k_1$  to obtain an intermediate ciphertext  $c'$ , and then encrypt  $c'$  using the Vigenère cipher with  $k_2$  to obtain the final ciphertext  $c$ . Show how to break the double Vigenère cipher.

**Solution** Let

$$k_1 = (k_{1,1}, \dots, k_{1,t_1}) \quad \text{and} \quad k_2 = (k_{2,1}, \dots, k_{2,t_2}),$$

where every entry is a shift in  $\mathbb{Z}_{26}$ . Set  $L = \text{lcm}(t_1, t_2)$  and define a length- $L$  key  $k = (k'_1, \dots, k'_L)$  by

$$k'_i = k_{1,((i-1) \bmod t_1)+1} + k_{2,((i-1) \bmod t_2)+1} \pmod{26}.$$

At position  $i$ , double encryption gives

$$\begin{aligned} c_i &= (m_i + k_{1,((i-1) \bmod t_1)+1}) + k_{2,((i-1) \bmod t_2)+1} \pmod{26} \\ &= m_i + k'_i \pmod{26}. \end{aligned}$$

Therefore, the double Vigenère cipher is a Vigenère cipher with key  $k$  and period  $L$ . (The minimum period of  $k$  may be smaller than  $L$ , but  $L$  is always a valid period.) We can consequently break the double cipher by applying the usual attack against a single Vigenère cipher.

## 3. Chosen Plaintext Attacks (10 points)

Assume the adversary has the ability to obtain ciphertexts for arbitrary plaintexts, i.e., it can choose a message and receive an encryption of the message without knowing the secret key. Attacks that leverage this information are known as *chosen plaintext attacks*. Using chosen plaintext attacks, show how to learn the secret key for the shift, substitution, and Vigenère ciphers. Your attacks should each use only a single plaintext. What is the smallest plaintext length that suffices for each attack?

For the Vigenère cipher, please consider two cases: (a) when the period  $t$  is known; (b) when the period  $t$  is unknown but an upper bound  $t_{max}$  on  $t$  is known. For the latter case, an asymptotic analysis of the required plaintext length suffices, i.e., an upper bound on the required plaintext length.

**Solution**

**Shift Cipher.** Choose an arbitrary message  $m \in \mathbb{Z}_{26}$  and request for the ciphertext  $c$  corresponding to  $m$ . Recover the key  $k$  as  $c - m \pmod{26}$ . One character is sufficient.

**Substitution Cipher.** For each  $m \in \{0, \dots, 24\}$ , request for the ciphertext encrypting  $m$ . Recover the permutation  $f$  of  $\mathbb{Z}_{26}$  that is consistent with  $\{i, c_i\}_{i \in \{0, \dots, 24\}}$  (i.e.,  $f(i) = c_i$ ) where  $c_i$  is the ciphertext corresponding to  $i$ . 25 characters are sufficient.

**Vigenère cipher of known period.** Let  $t$  be the period of the cipher. Choose an arbitrary message block of length  $t$  denoted by  $m_1 \dots m_t$  and obtain the ciphertext  $c_1 \dots c_t$  corresponding to this block. Recover  $k_\tau = c_\tau - m_\tau$  for each  $\tau \in [t]$ .  $t$  characters are sufficient.

**Vigenère cipher of unknown period.** Let  $T = t_{\max}$ . From a plaintext character  $m_i$  and its corresponding ciphertext character  $c_i$ , we can recover the shift used at position  $i$ :

$$s_i = c_i - m_i \pmod{26}.$$

Thus, a chosen plaintext of length  $T$  reveals the first  $T$  symbols  $s_1, \dots, s_T$  of the repeating key stream. However, these  $T$  symbols are not always enough to determine the period. For example, consider the following two keys (where  $0^j$  denotes a string of  $j$  zeros):

$$K = (0^{T-2}, 1, 0) \quad \text{and} \quad K' = (0^{T-2}, 1).$$

The first key has period  $T$ , while the second has period  $T - 1$ . Nevertheless, the first  $T$  key-stream symbols produced by both keys are

$$0^{T-2}, 1, 0.$$

For  $K'$ , the last zero is simply the first symbol appearing again as the key cycles. Thus, a length- $T$  observation cannot distinguish these two periods.

Intuitively, we want to recover enough of the key stream to see the key cycle back. Even then, the recovered sequence may have several candidate periods. For example, the sequence

$$0101010101$$

is consistent with periods 2, 4, and 6. The key idea is that there is a smallest candidate period, and every other candidate is a multiple of it.

**Lemma.** Let  $s$  be a string. If there is a repeating sequence of length  $t$  in  $s$  and there is a repeating sequence of length  $t'$  in  $s$ , then there is a repeating sequence of length  $\gcd(t, t')$  in  $s$ .

**Proof.** Assume without loss of generality that  $t' \geq t$ , and write

$$t' = qt + r, \quad 0 \leq r < t.$$

If  $r = 0$ , then  $t$  divides  $t'$  and  $t = \gcd(t, t')$ , so we are done. Suppose now that  $r > 0$ .

Write the length- $t$  repeating sequence of shifts as  $(x_1, x_2, \dots, x_t)$ . Since  $t' = qt + r$ , the first  $t'$  shifts consist of  $q$  complete copies of this sequence followed by its first  $r$  shifts:

$$\underbrace{\underbrace{(x_1, \dots, x_t) \cdots (x_1, \dots, x_t)}_{q \text{ copies}} (x_1, \dots, x_r)}_{t' = qt + r} \quad \left| \begin{array}{l} x_{r+1}, x_{r+2}, \dots \quad \text{according to the length-}t \text{ repetition,} \\ x_1, x_2, \dots \quad \text{according to the length-}t' \text{ repetition.} \end{array} \right.$$

The vertical line marks the end of the first length- $t'$  block. According to the length- $t$  repetition, we are  $r$  positions into a length- $t$  block at this point. Therefore, the next shifts must be  $x_{r+1}, x_{r+2}, \dots$ . On the other hand, according to the length- $t'$  repetition, a new length- $t'$  block begins at the vertical line. Therefore, the next shifts must instead be  $x_1, x_2, \dots$ .

These are two descriptions of the same shifts, so

$$x_1 = x_{r+1}, \quad x_2 = x_{r+2}, \quad \dots$$

where the indices wrap around after  $t$ . More generally,  $x_i = x_{i+r}$  for every  $i$ , with indices taken modulo  $t$ . Thus, shifting by  $r$  does not change the sequence, which means that there is also a repeating sequence of length  $r$  in  $s$ .

We can now repeat the same argument with  $t$  and  $r$ . This produces the next remainder in Euclid's algorithm. Continuing in this way, the final nonzero remainder is  $\gcd(t, t')$ . Hence there is a repeating sequence of length  $\gcd(t, t')$  in  $s$ .

**Example.** It may help to draw the argument for  $t' = 8$  and  $t = 5$ . Here  $T = 8$ , so all  $2T = 16$  shifts fit on one line. Start with the length-5 sequence  $(x_1, x_2, x_3, x_4, x_5)$ , treating its shifts as potentially different. Repeating this sequence gives

$$\underbrace{x_1, x_2, x_3, x_4, x_5, x_1, x_2, x_3}_{\text{first length-8 block}} \mid x_4, x_5, x_1, x_2, x_3, x_4, x_5, x_1.$$

Because the sequence also repeats every 8 positions, everything after the vertical line must equal the first length-8 block. Comparing corresponding positions shows that  $x_{i+3} = x_i$ . In particular,  $x_4 = x_1$  and  $x_5 = x_2$ . After replacing  $x_4$  by  $x_1$  and  $x_5$  by  $x_2$ , the original length-5 sequence becomes

$$(x_1, x_2, x_3, x_1, x_2),$$

which is generated by repeating the length-3 sequence  $(x_1, x_2, x_3)$ . This is the remainder step  $8 = 5 + 3$ .

We now redraw all 16 shifts using this new length-3 repetition:

$$\underbrace{x_1, x_2, x_3, x_1, x_2}_{\text{first length-5 block}} \mid x_3, x_1, x_2, x_3, x_1, x_2, x_3, x_1, x_2, x_3, x_1.$$

Because the sequence also repeats every 5 positions, everything after the vertical line must equal the first length-5 block. This gives  $x_{i+2} = x_i$ . In particular,  $x_3 = x_1$ . After replacing  $x_3$  by  $x_1$ , the length-3 sequence becomes

$$(x_1, x_2, x_1),$$

which is generated by repeating the length-2 sequence  $(x_1, x_2)$ . This is the remainder step  $5 = 3 + 2$ .

We redraw all 16 shifts once more using the new length-2 repetition:

$$\underbrace{x_1, x_2, x_1}_{\text{first length-3 block}} \mid x_2, x_1, x_2, x_1, x_2, x_1, x_2, x_1, x_2, x_1, x_2, x_1, x_2, x_1, x_2.$$

Because the sequence also repeats every 3 positions, the shifts after the vertical line must equal the first length-3 block. This gives  $x_{i+1} = x_i$ , and in particular  $x_2 = x_1$ . The length-2 sequence therefore becomes  $(x_1, x_1)$ , which is generated by repeating the length-1 sequence  $(x_1)$ . Thus the final updated row is

$$x_1, x_1, x_1, x_1, x_1, x_1, x_1, x_1, x_1, x_1, x_1, x_1, x_1, x_1, x_1.$$

This is the final remainder step  $3 = 2 + 1$ . Altogether, these are exactly the steps of Euclid's algorithm:

$$8 = 1 \cdot 5 + 3, \quad 5 = 1 \cdot 3 + 2, \quad 3 = 1 \cdot 2 + 1.$$

Hence  $\gcd(8, 5) = 1$ .

We now give the attack. Choose a plaintext  $m_1 \dots m_{2T}$  of length  $2T$  and obtain its ciphertext  $c_1 \dots c_{2T}$ . For every  $i \in [2T]$ , compute the corresponding shift

$$s_i = c_i - m_i \pmod{26}.$$

Call  $p \in [T]$  a candidate period if

$$s_i = s_{i+p} \quad \text{for every } i \in [2T - p].$$

The true period is a candidate because it is at most  $T$ . Let  $p$  be the smallest candidate and output  $(s_1, \dots, s_p)$  as the key.

The length  $2T$  ensures that every candidate period of length at most  $T$  has enough room to repeat twice. To see why the attack works, let  $p'$  be any other candidate period. Applying the lemma to these repeating descriptions shows that  $\gcd(p, p')$  is also a candidate. Since  $p$  is the smallest candidate, we must have  $\gcd(p, p') = p$ , and hence  $p$  divides  $p'$ . In particular,  $p$  divides the true period. Therefore, the original key is obtained by repeating  $(s_1, \dots, s_p)$ . Assuming, as usual, that the period of a Vigenère key means its smallest period,  $p$  is exactly the true period. Hence a single chosen plaintext of length  $2t_{\max}$  is sufficient.

#### 4. (optional) Exploring Kerckhoff's Principle

In the lectures, we covered Kerckhoff's principle: cryptographic systems should be designed such that the adversary knows everything except the secret key. We will see how this principle manifests in the real world through a small exercise.

`privacytools.io` maintains a crowdsourced list of tools that adhere to high security and privacy standards. In particular, the following link<sup>1</sup> lists file encryption software. Select at least one of these tools and:

- (a) Try out the tool. Observe that encrypted data is unintelligible and that the key payload is random bits.
- (b) Confirm that the codebase is open source, i.e., all critical code is publicly auditable.

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<sup>1</sup><https://www.privacytools.io/secure-file-encryption>

- (c) Assert that standardized cryptography are used, i.e., the encryption algorithm itself is publicly known. List the cryptographic protocols used and how they are standardized (e.g., AES standardized by NIST).
- (d) Observe that a lost encryption key means lost data: there is no way to recover encrypted data without the encryption key. This is in contrast to password recovery, which is effectively key recovery because the underlying keys are derived from the password.